

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Applied Business Analytics Practical
Practical 1: Introduction to Business Analytics:

AIM : Describing the Distribution of a Variable:

- a. Obtain a dataset and calculate descriptive statistics (mean, median, mode, variance, etc.) for a specific variable of interest.
- b. Create visualizations (histograms, box plots) to depict the distribution of a variable and analyze its characteristics.

Function / Method	Purpose	Description
pd.read_csv()	Load Data	Reads CSV file and stores it in a DataFrame
df.columns.str.strip()	Data Cleaning	Removes extra spaces from column names
df['Sales']	Variable Selection	Selects the Sales column for analysis
mean()	Average	Calculates the average value of data
median()	Middle Value	Finds the central value in sorted data
mode()	Most Frequent	Returns the most repeated value
var()	Variance	Measures spread of data
std()	Standard Deviation	Shows dispersion from mean
max() - min()	Range	Calculates difference between highest and lowest value

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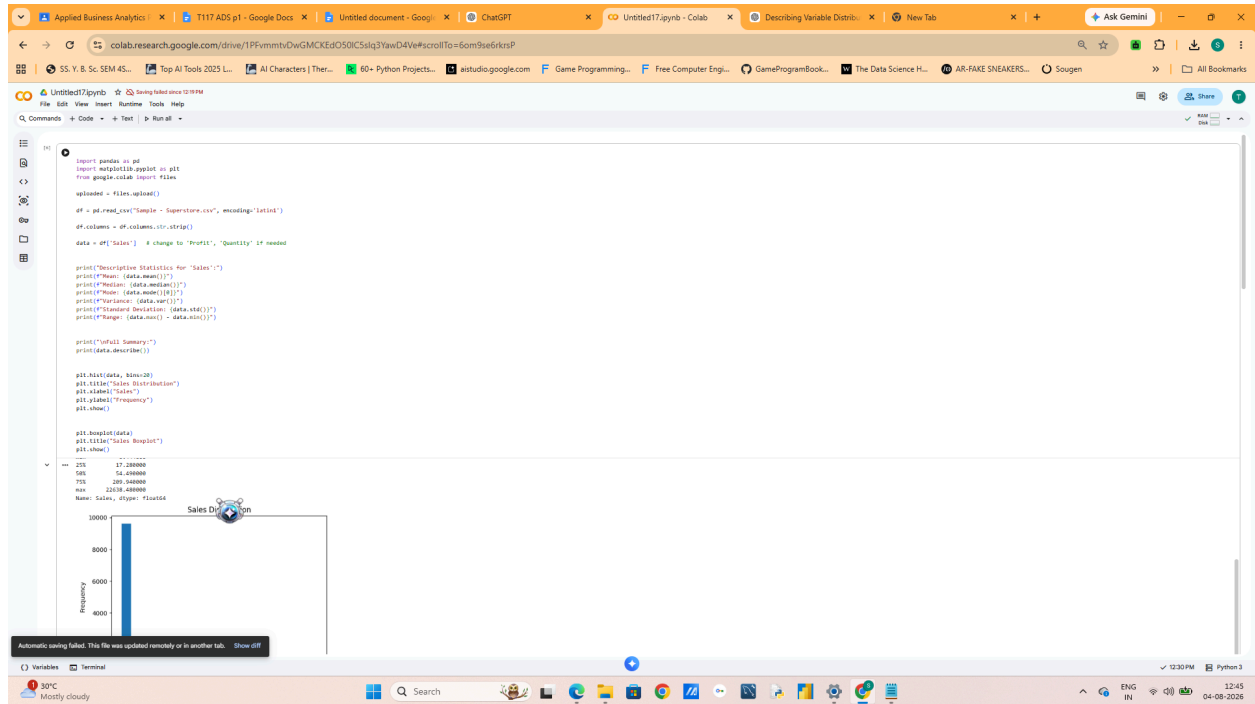
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Table – Visualization Functions

Function / Method	Purpose	Description
plt.hist()	Histogram	Displays frequency distribution of data
bins=20	Data Grouping	Divides data into intervals
plt.title()	Title	Adds title to the graph
plt.xlabel()	X-axis Label	Labels horizontal axis
plt.ylabel()	Y-axis Label	Labels vertical axis
plt.show()	Display	Shows the plotted graph
plt.boxplot()	Boxplot	Displays quartiles and spread of data

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Code :



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```
import pandas as pd
import matplotlib.pyplot as plt
from google.colab import files

uploaded = files.upload()

df = pd.read_csv("Sample - Superstore.csv", encoding='latin1')

df.columns = df.columns.str.strip()

data = df['Sales'] # change to 'Profit', 'Quantity' if needed

print("Descriptive Statistics for 'Sales':")
print(f"Mean: {data.mean()}")
print(f"Median: {data.median()}")
print(f"Mode: {data.mode()[0]}")
print(f"Variance: {data.var()}")
print(f"Standard Deviation: {data.std()}")
print(f"Range: {data.max() - data.min()}")

print("\nFull Summary:")
print(data.describe())

plt.hist(data, bins=20)
plt.title("Sales Distribution")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()

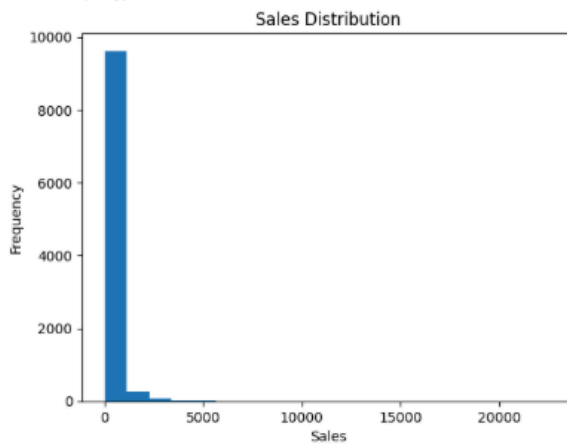
plt.boxplot(data)
plt.title("Sales Boxplot")
plt.show()
```

Output :

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```
** Choose Files Sample - Superstore.csv
Sample - Superstore.csv(text/csv) - 2287808 bytes, last modified: 8/4/2028 - 100% done
Saving Sample - Superstore.csv to Sample - Superstore.csv
Descriptive Statistics for 'Sales':
Mean: 229.8580003049833
Median: 54.489999999999995
Mode: 12.96
Variance: 388434.45530807687
Standard Deviation: 623.2451005086818
Range: 22638.036
```

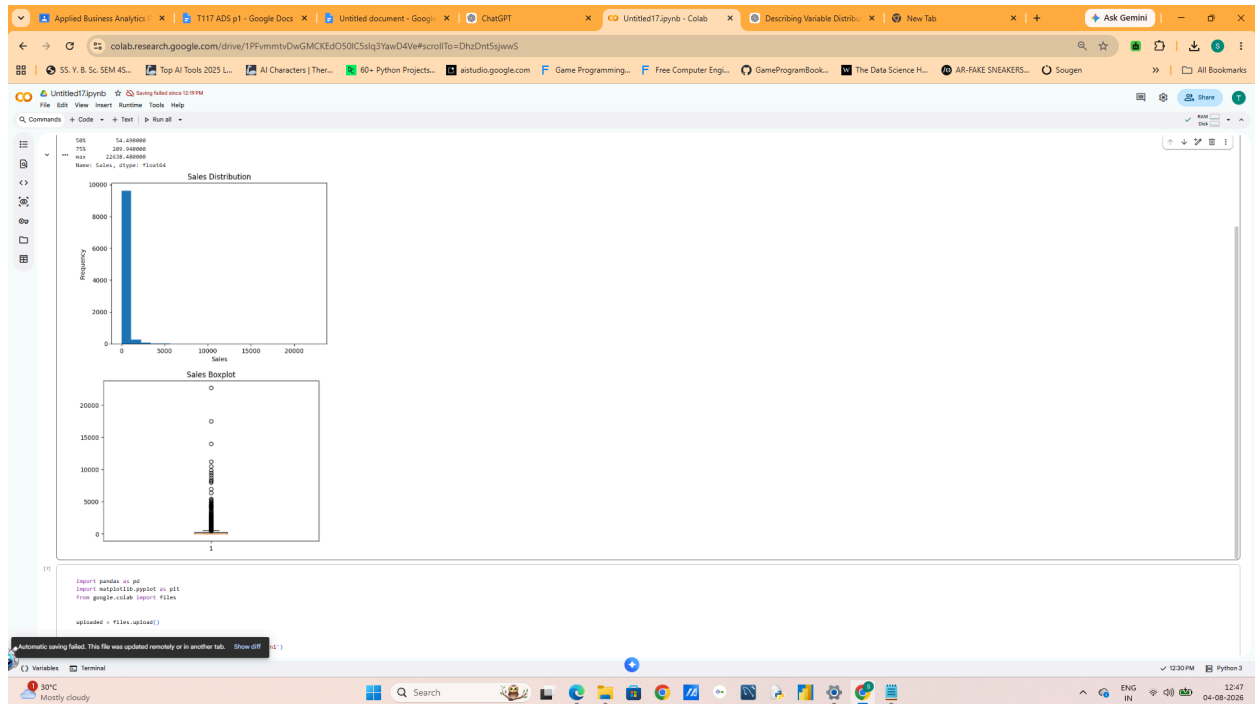
```
Full Summary:
count    9994.000000
mean     229.858001
std      623.245101
min       0.444000
25%      17.280000
50%      54.490000
75%     289.940000
max     22638.480000
Name: Sales, dtype: float64
```



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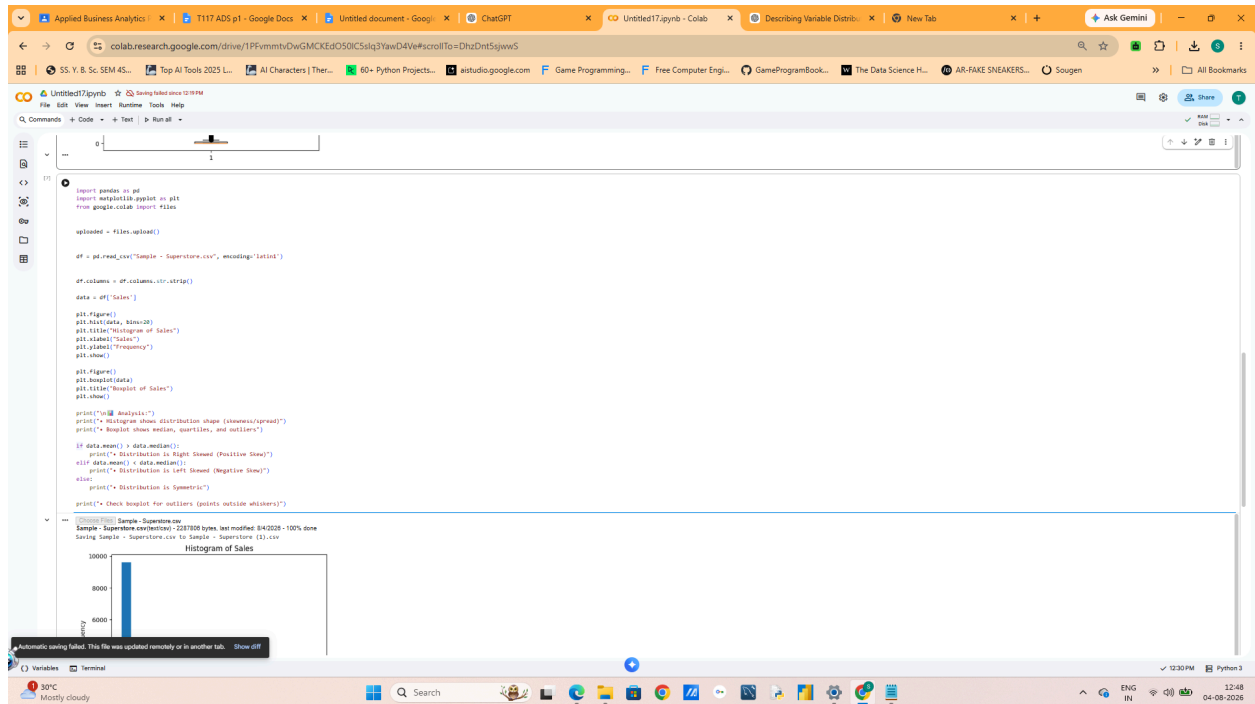
b. Create visualizations (histograms, box plots) to depict the distribution of a variable and analyze its characteristics.

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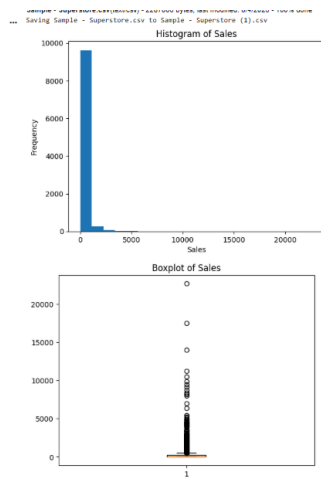
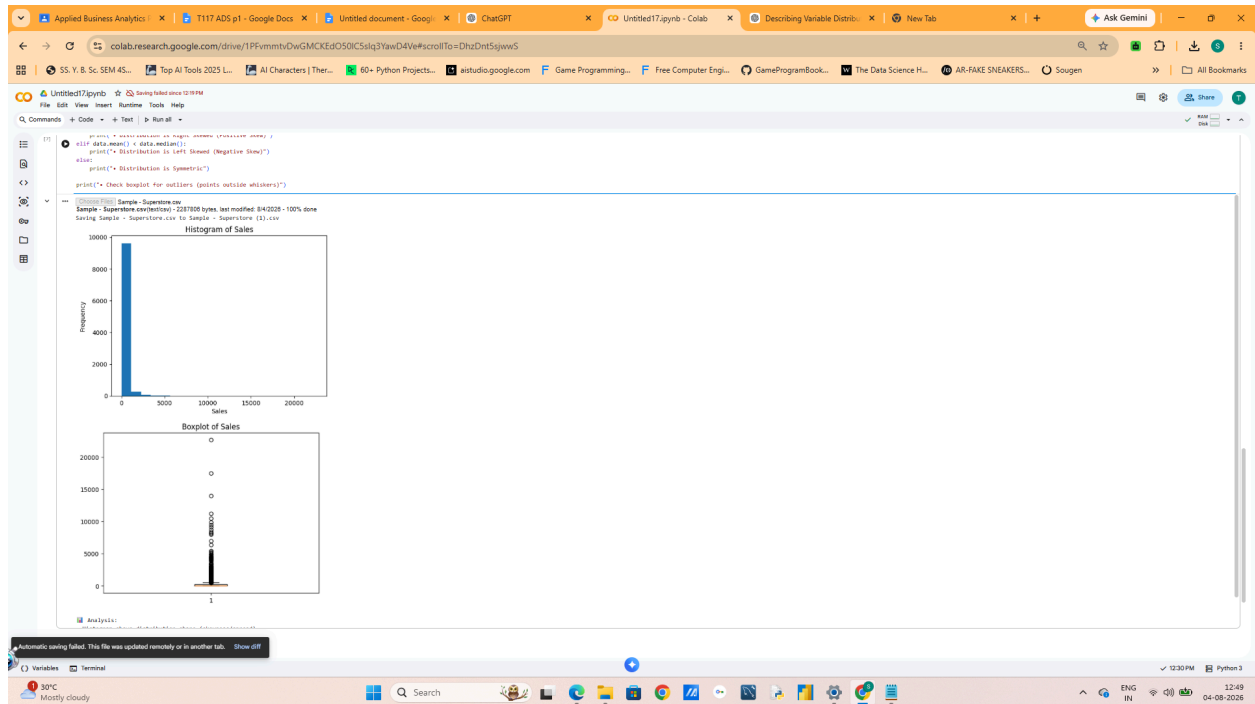
Output :

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